My 3 level SOC home Lab architecture

By Joseph Fansi (03.04.2026)

My Certifications:

IT

•Scrum Master
•Google Project Management & Agile
•IT Service Management

SOC

•ComptiaSec+
•Presecurity_TryHackMe
•Cyber Security 101_TryHackMe
•SOC Level 1 _TryHackMe
•Advent of Cyber 2025_ TryHackMe

SOC Experience:

•Daily use of SOC and Threat-hunting Simulators

•Home SOC Lab – Malware Simulation & Network Investigation:

✓ Built a cybersecurity lab
✓ Generated and executed malware
✓ Monitored attack activity
✓ Investigated large network captures
✓ Correlated host telemetry and network logs to detect indicators of compromise

1. Project Overview

Goal

Learn:

how logs are generated

how they move across a network

how a SOC detects something simple

Data Flow (Simple + Clear)

1.Attack happens from **Attacker VM Kali** (192.168.100.30)

2.Victim VM Windows logs events using:

- Sysmon

3.Logs go to **Ubuntu Server VM** SIEM (192.168.100.10)

4.SIEM (Wazuh) detects suspicious activity

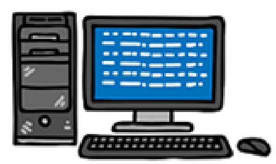
2. Beginner-Intermediary
SOC Lab Architecture
Flat Network (Controlled Visibility)

Cybersecurity community support:
Reddit, Github, LinkedIn, Youtube,
TryHackMe, Professionals, Cloud (SaaS)

AI support

Network:
Name_Easy SOC lab
Type: Host-Only Adapter
Subnet: 192.168.100.0/24

VM Network Config
Set ALL VMs:
Adapter 1 → Host-Only Adapter → SOC-LAB



Attacker VM (RDP / SMB / PowerShell)
Kali / Ubuntu
IP: 192.168.100.30

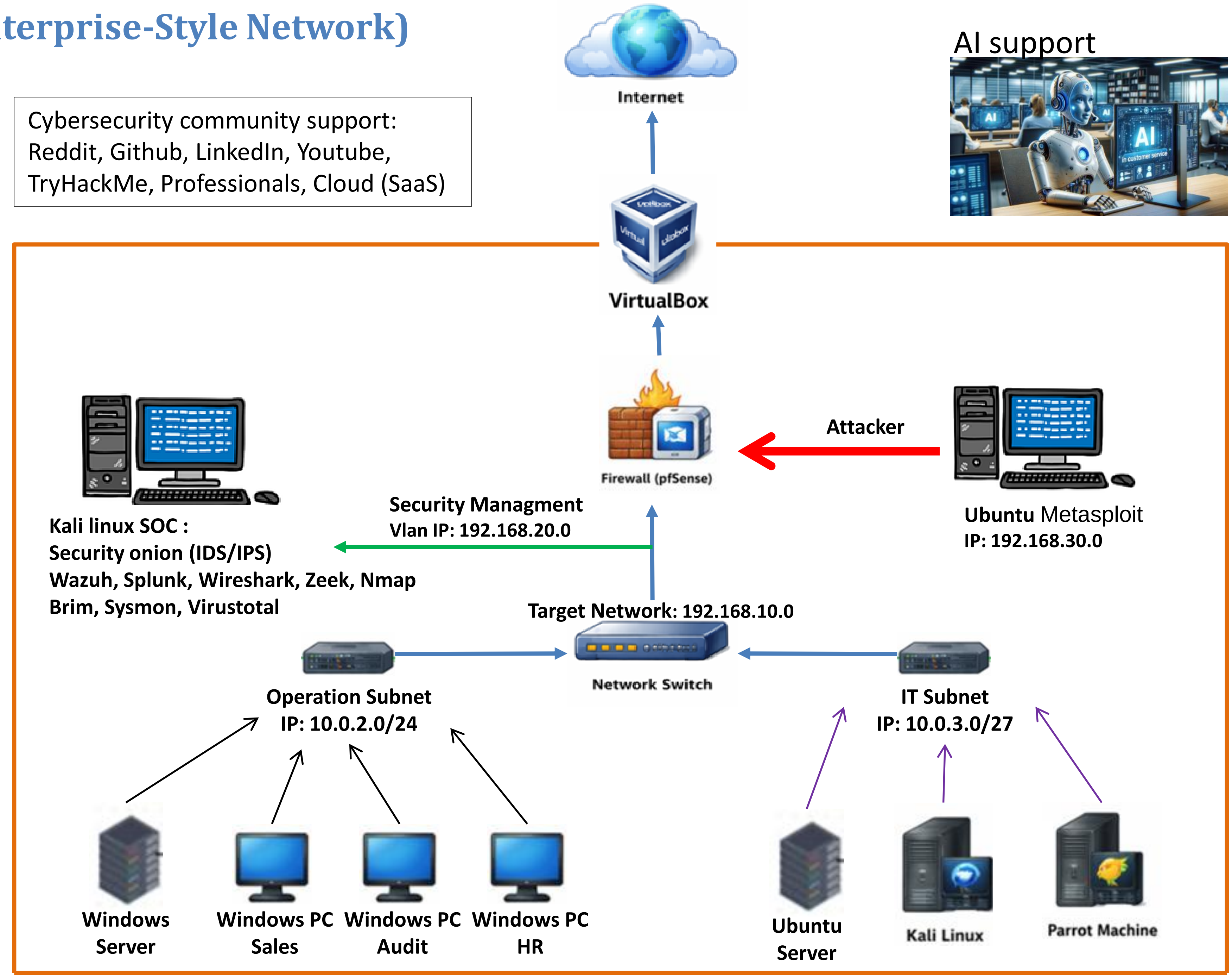
RDP / SMB / PowerShell

Victim VM (Sysmon logs)
Windows 10 PC
IP : 192.168.20.0

Log forwarding

Ubuntu Server VM: Log server
SOC-SIEM: Wazuh
IP: 192.168.100.10

3. Advanced SOC Lab Architecture (Enterprise-Style Network)



Thank you & good luck with your SOC Lab Project